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1 BUS 411 lecture topics

This map reflects the current set of 11 lecture notes.

1.1 Lecture 1: Introduction

- Bond cash-flow structures, maturity, principal, and coupon types
- Amortization and embedded options
- Identifying bond issues and financial innovation
- Fixed-income market sectors
- Interest-rate, reinvestment, call, credit, inflation, currency, liquidity, and volatility risk

1.2 Lecture 2: Bond Pricing and Yield Measures

- Future value, present value, annuities, and discounting cash-flow streams
- Pricing zero-coupon and coupon bonds between payment dates
- Clean price, full price, accrued interest, and quotation conventions
- Floating-rate and inverse-floating-rate securities
- Current yield, yield to maturity, yield to call, yield to put, yield to worst, and portfolio yield

1.3 Lecture 3: Bond Price Volatility, Duration, Convexity, and Bond Portfolio Strategies

- Price sensitivity to coupon, maturity, and yield
- PVBP, duration, spread duration, and portfolio duration
- Analytical versus empirical duration
- Convexity and duration-convexity price approximations
- Yield-curve reshaping duration and key-rate duration

1.4 Lecture 4: Benchmark Spreads and the Term Structure of Interest Rates

- Drivers of benchmark spreads: credit, options, tax, liquidity, financing, and maturity
- Yield curves, spot rates, forward rates, and Treasury curve construction
- Expectations, liquidity preference, and market segmentation
- Bull and bear steepeners and flatteners
- Butterfly, carry, and roll-down strategies with risk controls

1.5 Lecture 5: Treasury and Corporate Debt Markets

- Treasury bills, notes, bonds, auctions, secondary trading, taxation, and STRIPS
- Agency and government-sponsored enterprise securities
- Corporate capital structure, seniority, bankruptcy, covenants, and ratings
- Corporate bonds, medium-term notes, commercial paper, and bank loans
- Default, recovery, downgrade, and credit-spread risk

1.6 Lecture 6: Bond Portfolio Construction

- Applying portfolio theory to bonds
- Passive, active, and benchmark-aware construction
- Tracking error calculation and interpretation
- Risk-factor, cell-matching, and optimization approaches
- Rebalancing and active-risk control

1.7 Lecture 7: Interest-Rate Models

- Short-rate processes, drift, volatility, and mean reversion
- Yield curves versus stochastic rate models
- Short-rate trees and future discount curves
- Hull–White model intuition and calibration
- Risk-neutral probabilities, one-period trees, and backward induction

1.8 Lecture 8: Valuation of Bonds with Embedded Options and Option-Adjusted Measures

- Limits of traditional yield-spread analysis and static spread
- Callable and puttable bond economics
- Option-free versus embedded-option value decomposition
- Binomial interest-rate trees and backward-induction valuation
- Option-adjusted spread, effective duration, and effective convexity

1.9 Lecture 9: Interest-Rate Futures Contracts

- Futures mechanics, margining, and marking to market
- Futures versus forwards and payoff characteristics
- Treasury and short-term interest-rate futures
- Carry, arbitrage, conversion factors, and cheapest-to-deliver analysis
- Duration/DV01 hedging and portfolio applications

1.10 Lecture 10: Credit Spread Exposure and Credit Risk Modeling

- Absolute and relative spread changes, spread duration, and DTS
- Empirical duration and duration multipliers for corporate bonds
- Credit ratings and the challenges of modeling default risk
- Structural and reduced-form credit models
- Default dependence, copulas, portfolio credit risk, and relative-value applications

1.11 Lecture 11: Interest-Rate Options

- Calls, puts, exercise, assignment, quotation, premium, and margin conventions
- Futures options versus futures and the yield-to-price-to-option connection
- Intrinsic value, time value, profit, break-even prices, and put-call parity
- Bond-option valuation with Black-Scholes-style formulas and binomial trees
- Futures-option valuation with Black's model
- Treasury futures option applications, including asymmetric protection and income strategies